AMRITA VISHWA VIDYAPEETHAM
DEEMED TO BE UNIVERSITY UNDER SECTION 3 OF UGC ACT, 1956
COIMBATORE CAMPUS

INSTITUTION'S INNOVATION COUNCIL
(Ministry of Education Initiative)

GENESIS

Genesis Ideathon: Final Presentation

Team ID: *GEN26-A0H-I052*

Team Name: *CODEWAVE*

Domain: *Intelligent Systems and Digital Futures*

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Date: 11th October 2026

Presentation Outline

1. Problem Statement
2. Problem Understanding
3. Existing Gaps & Challenges
4. Our Proposed Solution
5. Technical Approach
6. Innovation & Uniqueness
7. Feasibility & Scalability
8. Cost for Product
9. Use Cases
10. Use Cases
11. Use Cases
12. Impacts and Benefits

Problem Statement

How can we prevent AI assistance from becoming cognitive dependency?
The growing reliance on AI-generated direct answers which results in reduction of opportunities for learners to practice independent reasoning, critical thinking, and problem-solving.
The challenge is to make AI a tool that strengthens human thinking rather than routinely replacing it.

Problem Understanding

- Who experiences the problem?
  - Students and learners who regularly use generative AI.
  - Individuals using AI for coding, writing, research, and problem-solving.

- What causes the problem?
  - A shift from beneficiary scaffolding to detrimental scaffolding.
  - AI gives abrupt answers without allowing the user to think.
  - Limited emphasis on independent attempts and knowledge transfer.

- How are users affected?
  - Decline in human cognitive skills in the form of critical thinking and problems solving.
  - Weakening of neuroplasticity and atrophy which leads to synaptic pruning due to metacognitive laziness
  - It creates a phase where human cannot solve problems without AI assistance

Existing Gaps & Challenges

- Existing Gap - 1: Many interventions are short-term or pilot studies; long-term efficacy and transfer to real-world settings remain uncertain.
- Existing Gap - 2: Many AI interactions prioritize task completion, even when the user needs to develop a skill.
- Main challenge: A challenge that remains unsolved is that AI can bring about solutions but can it deliver in a way that the user learns the required skill ?

Our Proposed Solution

- Core idea: Our idea is to create a middleware that works between the user and the AI chatbot that adapts to the learning demands of the user.

- How the solution works at a high level?
  - It acts as a socratic tutor which prompts the user to think about the problem instead of giving the solution directly.
  -Task Aware Assistance:
  Level 0 — Direct Assistance
  Level 1 — Cognitive Prompting
  Level 2 — Progressive Scaffolding
  Level 3 — Independent Transfer

- After the interaction with AI, the user will be able to:
  -Interpret the concept and solve similar problems without AI assistance.

Technical Approach

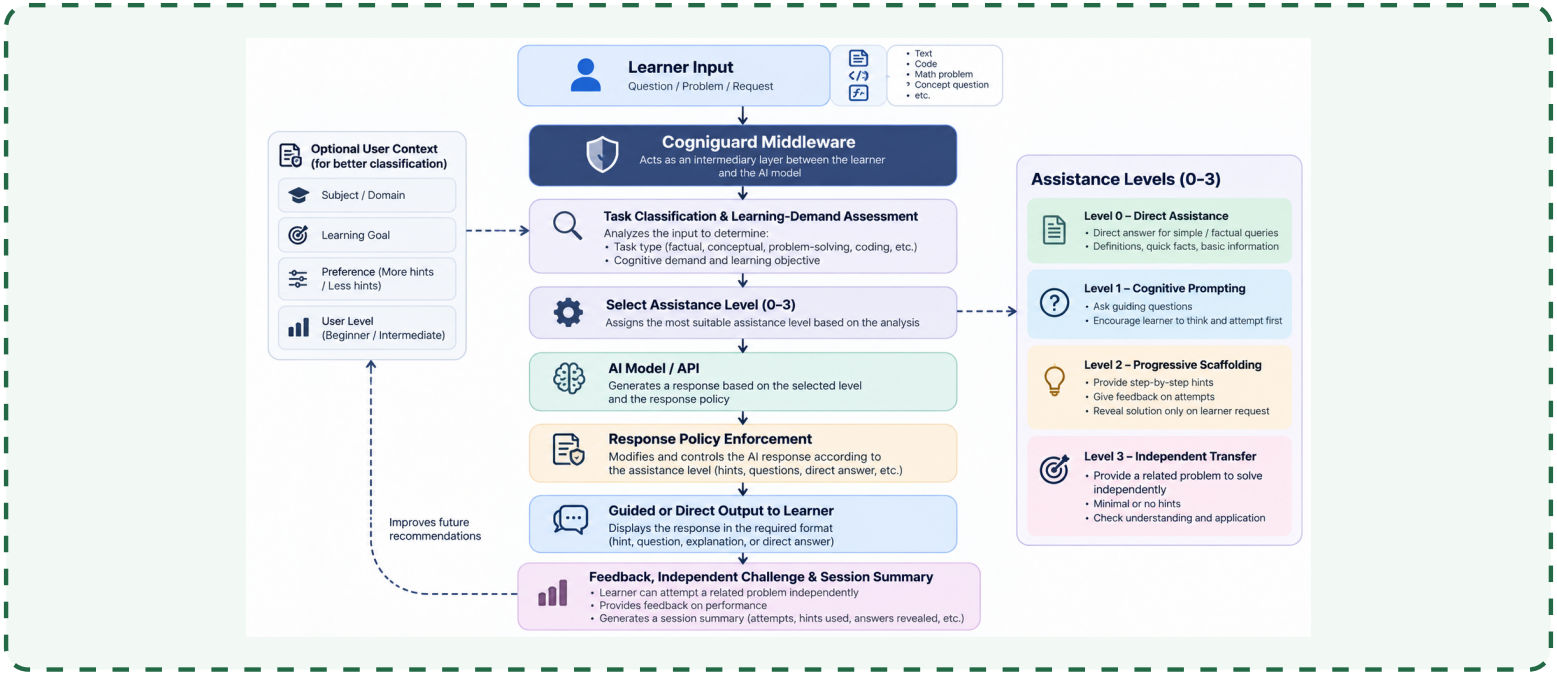


Figure: Proposed Cogniguard middleware architecture.

The middleware evaluates the request, selects an assistance policy, and coordinates with the AI model. The resulting response is delivered according to the selected level. Session data can record attempts, hints, solution reveals, and independent-challenge performance.

Tech Stack

Frontend: Middleware

Database: Python with FastAPI

Backend: SQLite for an initial prototype; a scal-

AI/ML, APIs, or Hardware: External LLM API, task-classification logic, and response-policy en-

able database can be

forcement

Innovation & Uniqueness

- What is new or different?
  - Progressive hints
  - Redirecting the user back to track if they go wrong
  - Evaluating the user response in every step
  - Providing downloadable session summaries for further reference

- Why is this approach suited to the problem?
  Learners need different amounts of support for different tasks. A factual question does not necessarily need the same intervention as a multi-step reasoning problem.

- What is the key differentiator?
  - Task segregation according to the level of the problem
  - It acts as a socratic tutor rather than a traditional chatbot

Feasibility & Scalability

- **Feasibility:**
  - Begin with a controlled web-based prototype.
  - Implement four assistance levels using predefined rules.
  - Integrate an AI API and response-policy layer incrementally.
  - Validate the interaction with students and educator

- **Scalability:**
  - It extends help for academic learners and working professionals
  - Add personalized assistance policies and progress tracking
  - Evaluate effectiveness across different learner groups

- **Risks:**
  - Privacy concerns: Minimize stored data and obtain appropriate consent.
  - Task misclassification: Use confidence thresholds and allow correction or fallback.
  - Unproven learning outcomes: Conduct controlled evaluations before making effectiveness claims.

Cost for Product (Optional)

- Development or hardware cost: Student-built using existing computers and free development tools.
- Estimated total: Actual operating costs require provider-specific quotations and usage estimates. Do not present an unverified cost as a confirmed budget.

Use Cases (compulsorily 3 use cases must be there)

- Use case 1: Calculus Learning

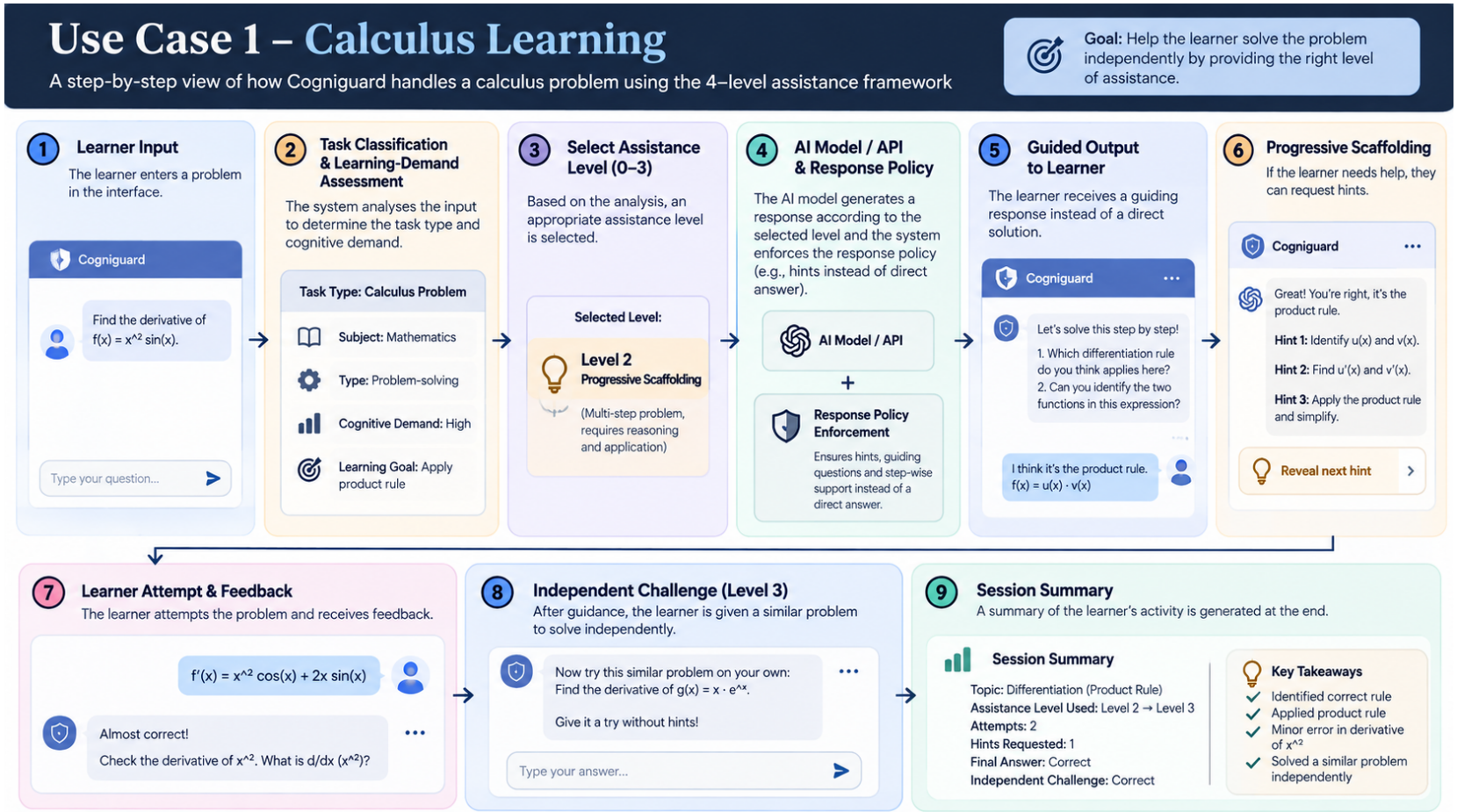


Figure: Guided calculus learning through progressive hints and independent transfer. (AI-generated illustrative visual of the proposed interaction.)

Use Cases (compulsorily 3 use cases must be there)

- Use case 2: Introductory Python

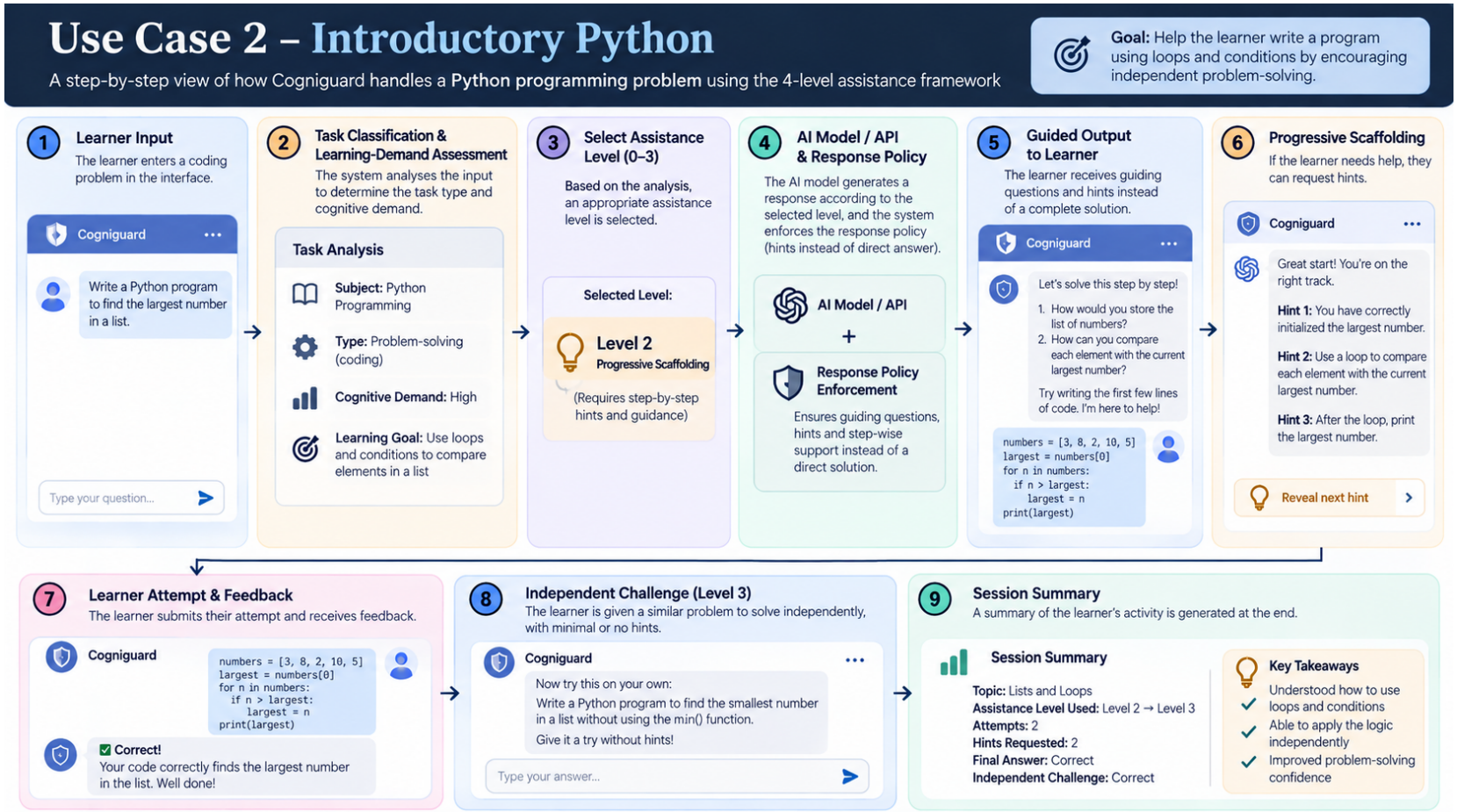


Figure: Guided Python learning with structured feedback and an independent coding challenge. (AI-generated illustrative visual of the proposed interaction.)

- Use case 3: Direct Factual Assistance

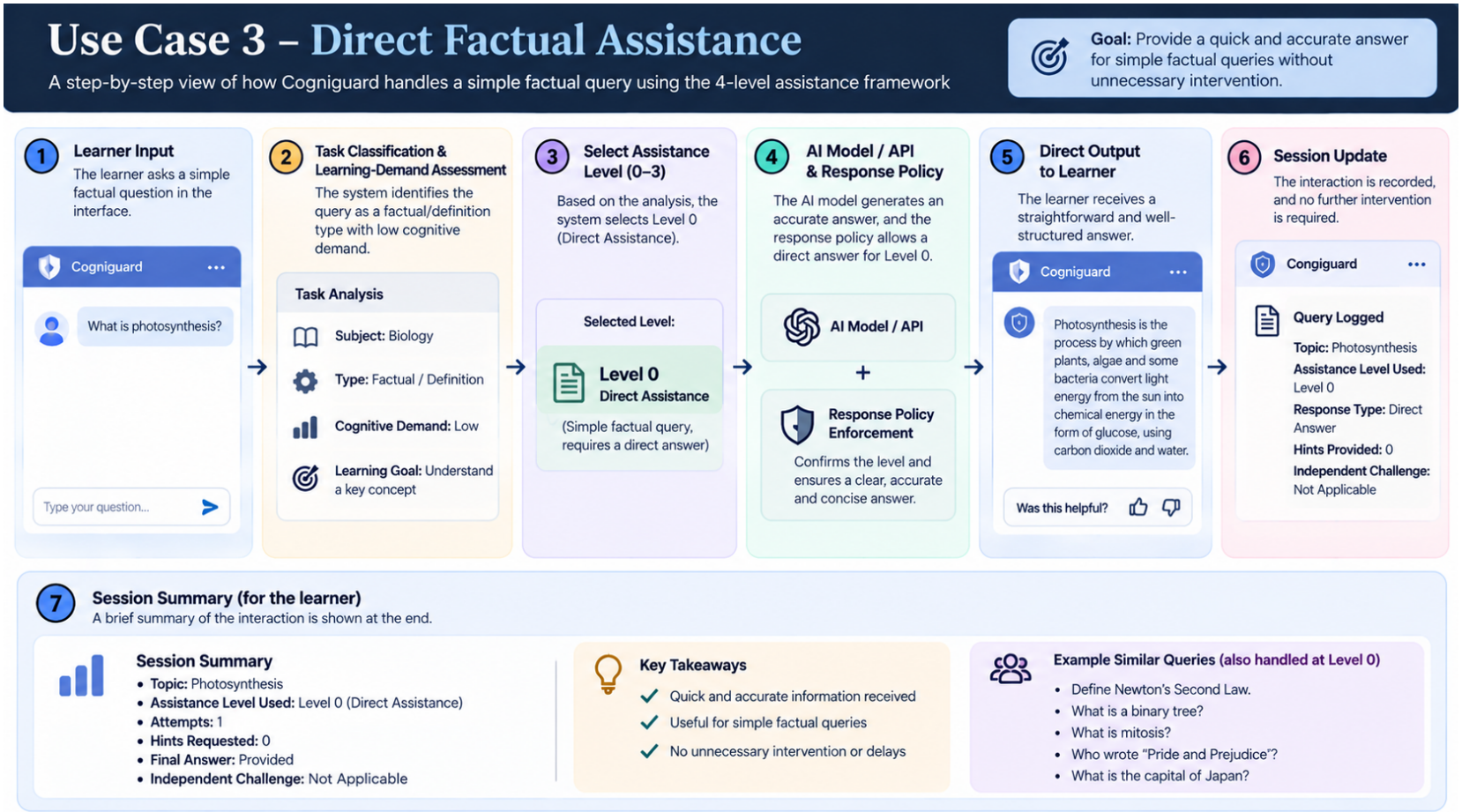


Figure: Level 0 direct assistance for simple factual queries. (AI-generated illustrative visual of the proposed interaction.)

Impacts and Benefits

- Benefit to the target users:
  - It provides platform for the user to learn the concept rather than solving a single problem.

  Benefits to educators and institutions:
  - Potential support for learner-centred teaching strategies.

- Expected measurable outcome:
  - Accuracy on independent transfer tasks.
  - Number of attempts and hints required.
  - Frequency of worked-solution reveals.
  - Performance across repeated, comparable tasks.

Sustainable Development Goals (SDGs)

SDG Number	Target Number	Goal Name	Connection
SDG 4	Target 4.4	Quality Education	Cogniguard aims to support independent problem-solving and responsible use of AI in learning.
SDG 9	Target 9.5	Industry, Innovation and Infrastructure	Cogniguard proposes an innovative software layer for human-centred AI interaction and provides a basis for further research and evaluation.

Individual Member Contribution

Member	Contribution or responsibility
SAYANTIKA DAS	Team leadership and coordination, Product strategy and problem analysis, Presentation editing and coordination, Prototype development support
DEBANGSHI BHATTACHARYA	Enhanced ideas through suggestions and critical questioning, Contributed to problem analysis, Led chart-paper design and aesthetic development
SEEVAANEE E	Proposed the middleware-based solution, Led research and evidence verification, Collaborated on product strategy and problem analysis
KOLLAPANENI HIMANI	Contributed to problem analysis and idea generation, Collaborated in problem analysis and research, Led physical chart-paper writing and preparation

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